

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX)		**Date:20-12-24**
Time: 3HRS	**GTM-2**	**Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. Two polarizers are oriented with transmission planes making an angle of $60°$. The percentage of incident unpolarized light that passes through the system is

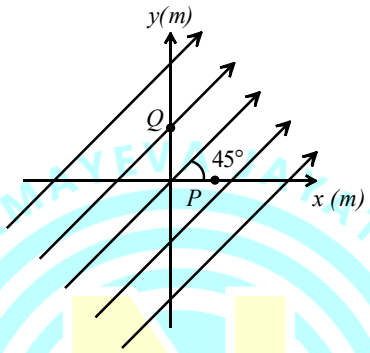
   A) 50% B) 100% C) 12.5% D) 37.5%

2. If a rubber ball is taken at the depth of $200\,m$ in a pool, its volume decreases by 0.1%. If the density of the water is $1\times10^{3}\,kg/m^{3}$ and $g=10\,m/s^{2}$, then the volume elasticity in N/m^{2} will be:

   A) 10^{8} B) 2×10^{8} C) 10^{9} D) 2×10^{9}

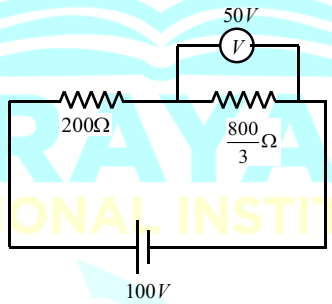
For More Material Join: @JEEAdvanced_2025

3. A uniform electric field of $200V/m$ is directed at $45°$ with $x-$ axis as shown in figure. The coordinates of point P and point Q are $(1,0)$ and $(0,2)$. Find the potential difference $V_P - V_Q$ (in volts).



A) $\frac{200}{\sqrt{2}}$ B) $200\sqrt{2}$ C) $\frac{-200}{\sqrt{2}}$ D) $-200\sqrt{2}$

4. What is the resistance of voltmeter shown in the circuit?



A) $\frac{800}{3}\Omega$ B) 800Ω C) 1000Ω D) 200Ω

5. A steel wire of diameter $2\ mm$ has a breaking strength of $4\times10^5\ N$. What is the breaking force of similar steel wire of diameter $1.5\ mm$?

A) $2.25\times10^5\ N$ B) $2.6\times10^5\ N$ C) $3\times10^5\ N$ D) $1.5\times10^5\ N$

6. A car runs at constant speed on a circular track of radius $100\,m$ taking $62.8\ s$ on each lap. What is the average speed and average velocity on each complete lap?

A) Velocity $10\ m/s$ speed $10\ m/s$ B) Velocity zero, speed $10\ m/s$

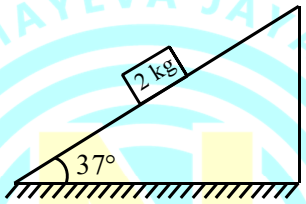
C) Velocity zero, speed zero D) velocity $10\,m/s$, speed zero

For More Material Join: @JEEAdvanced_2025

7. The deflection in a galvanometer falls from 50 divisions to 20 divisions, when a 12Ω shunt (resistance connected in parallel to galvanometer) is applied. The galvanometer resistance is

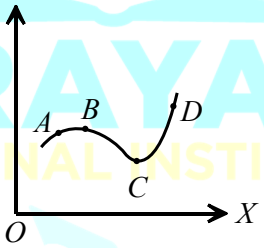
A) 18Ω B) 24Ω C) 30Ω D) 36Ω

8. A block of mass $2kg$ is at rest on an inclined plane of inclination angle $37°$. Calculate the force of friction acting on the block [take $g = 10\ m/s^2$]



A) 20N B) 12N C) 16N D) Zero

9. For a particle moving under the action of a variable force, kinetic energy (K) versus position (x) graph is given, then



A) At A, particle is decelerating

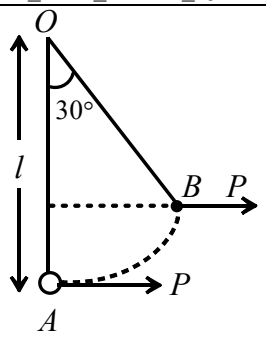
B) At B, particle is decelerating

C) At C, particle has maximum velocity

D) At D, particle has maximum acceleration

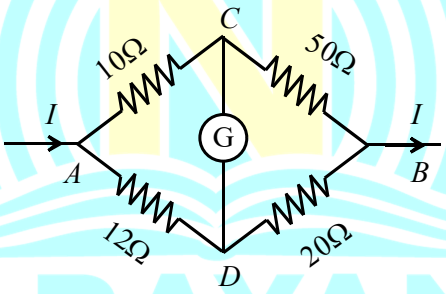
10. A bob of mass m is pulled along a circular arc by means of a constant horizontal force P as shown. Work done by the force P while lifting the bob from A to B is

For More Material Join: @JEEAdvanced_2025



A) mgl B) $\frac{Pl}{2}$ C) Pl D) $\frac{\sqrt{3}Pl}{2}$

11. In the situation shown, what is the value of resistance connected between A and D in addition to 12Ω resistance, so that galvanometer shows no deflection?



A) 12Ω B) 6Ω C) 4Ω D) 5Ω

12. Two vessels having equal volume contain molecular hydrogen at one atmosphere and helium at two atmospheres respectively. If both samples are at the same temperature, the rms velocity of hydrogen molecule is

A) Equal to that of helium B) Twice that of helium

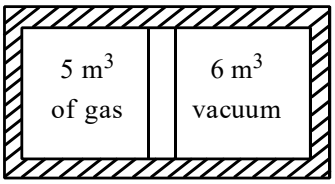
C) Half that of helium D) $\sqrt{2}$ times that of helium

13. A uniform ladder of length $3.25m$ and weighing $240N$ is placed against smooth vertical wall with its lower end $1.25m$ from the wall. If coefficient of friction is 0.3. Find the friction force between ladder and floor (Take normal reaction perpendicular to wall and floor)

A) 20 N B) 30N C) 50 N D) 40 N

For More Material Join: @JEEAdvanced_2025

14. A movable piston separates a sample of gas from vacuum. In this state, pressure of gas is P, then find pressure after piston is released.



A) P B) $6P$ C) $\frac{5P}{11}$

D) Pressure is not defined because it is a free expansion.

15. In a thermodynamic process on an ideal diatomic gas, work done by the gas is η times the heat supplied $(\eta < 1)$. The molar heat capacity of the gas for the process is

A) $\frac{5R}{2}$ B) $5\eta\frac{R}{2}$ C) $\frac{5R}{2(1-\eta)}$ D) $\frac{5R(1-\eta)}{2}$

16. **Assertion:** The time period of revolution of a satellite close to surface of earth is smaller then that revolving away from surface of earth.
Reason: The square of time period of revolution of a satellite is directly proportional to cube of its orbital radius.

A) If both assertion and reason are true and reason is the correct explanation of assertion.

B) If both assertion and reason are true but reason is not the correct explanation of assertion

C) If assertion is true but reason is false

D) If both assertion and reason are false

17. **Assertion (A):** The accuracy of a standard screw gauge is more than the accuracy of a standard vernier caliper.

Reason (R) : The least count of standard screw gauge is more than the least count of standard vernier caliper.

A) A and R are true and R is the correct explanation of A

B) A and R are true, but R is not the correct explanation of A

C) A is true but R is false

D) Both A and R are false

For More Material Join: @JEEAdvanced_2025

18. Given that $y = a\sin\left(\frac{t}{p} - qx\right)$, where t represents time in second and x represents distance in meter. Which of the following statement is true?

A) Dimension of t is same as that of q

B) Dimension of x is same as that of p

C) Dimension of x is same as that of q

D) Dimension of t is same as that of p

19. A nucleus disintegrates into two parts which have their velocities in the ratio 2 : 1 The ratio of their nuclear sizes will be

A) $2^{1/3}:1$ B) $1:3^{1/2}$ C) $3^{1/2}:1$ D) $1:2^{1/3}$

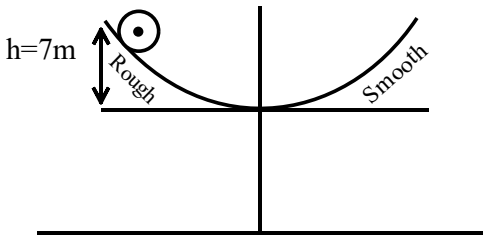
20. In a photo-emissive cell, with exciting wavelength λ_1, the maximum kinetic energy of electron is K. If the exciting wavelength is changed to λ_2, the maximum kinetic energy of electron is $2K$ then

A) $\lambda_1 = 2\lambda_2$ B) $\lambda_1 > 2\lambda_2$ C) $\lambda_1 < 2\lambda_2$ D) $\lambda_1 = \frac{\lambda_2}{2}$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

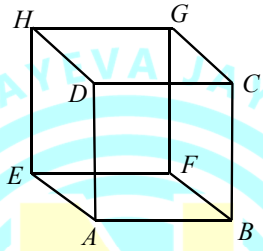
This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. A solid sphere is released from rest on a curved surface as shown in figure. Left half of the surface is rough and right half is smooth. If sphere is released from rough side, then the maximum height attained by it on smooth side is (Rough surface has sufficient friction for pure rolling)

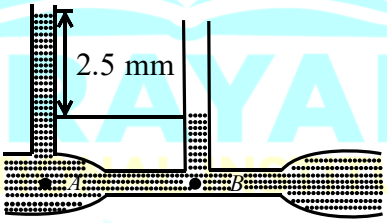


For More Material Join: @JEEAdvanced_2025

22. Consider an imaginary cube with different vertices of the cube marked as shown. A point charge is placed at vertex H and flux through the face $ABCD$ is ϕ in this case. Now, the charge is shifted to centre of the cube. Flux through the face $AEHD$ in this case is $k\phi$. What is the value of k ?

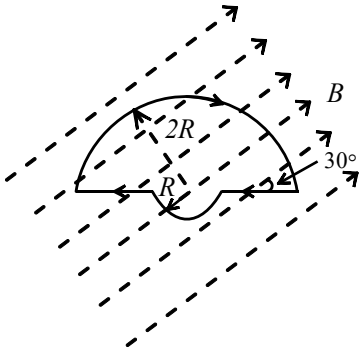


23. In the figure shown below, a fluid of density $2\times10^{3} kg\ m^{-3}$ is flowing in a horizontal pipe. The speed of water at point A is $4 cm\ s^{-1}$, what is its speed at point B (in $cm\ s^{-1}$). Round off the answer to the nearest integer.



24. In a stationary wave represented by $y = a\sin\omega t\cos kx$, amplitude of the component progressive wave is na. The value of $10n$ _____

25. The torque acting upon the following structure carrying current i due to the magnetic field B is $\frac{p}{q}\pi R^2 iB$. Find $p+q$. (Magnetic field is in the plane of loop)



For More Material Join: @JEEAdvanced_2025

CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. For the homogeneous 1st order gaseous reaction.

$A \rightarrow B+C+D+E+F+G+H+I+J+K+L+M+N+O+P+Q+R+S+T+U+V+W+X+Y+Z$

The initial pressure was P_0 while after time t, total pressure was P. The expression for rate constant k in terms of P_0, P and t is

A) $k=\frac{2.303}{t}\log\frac{P_0}{P_0-P}$ B) $k=\frac{2.303}{t}\log\frac{25P_0}{26P_0-P}$

C) $k=\frac{2.303}{t}\log\frac{24P_0}{25P_0-P}$ D) $k=\frac{2.303}{t}\log\frac{24P_0}{P_0-25P}$

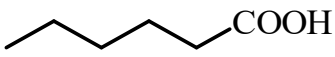
27. In the following reaction, the number of monochloro derivatives possible for the compound 'D' is

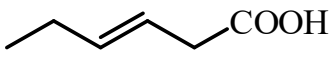
$$CH_3CN \xrightarrow{H_2O/H^+} A \xrightarrow{Ca} B \xrightarrow[\Delta]{\text{dry distillabtion}} C \xrightarrow{Zn - Hg + HCl} D$$

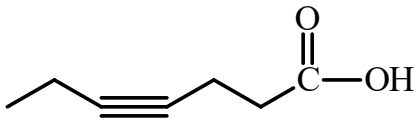
A) 3 B) 4 C) 2 D) 5

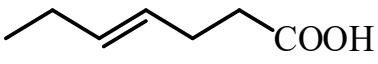
28. What is product of following reaction?

$$\text{Hex—3—ynal} \xrightarrow[\text{(ii) } PBr_3]{\text{(i) } NaBH_4} \xrightarrow[\text{(iv) } CO_2/H_3O^+]{\text{(iii) Mg/ether}}$$

A) 

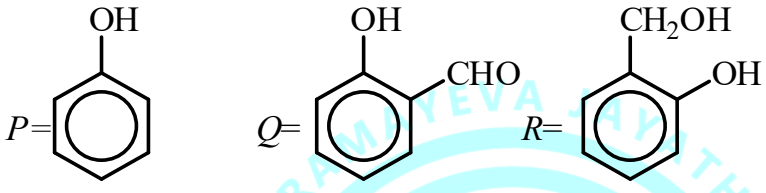
B) 

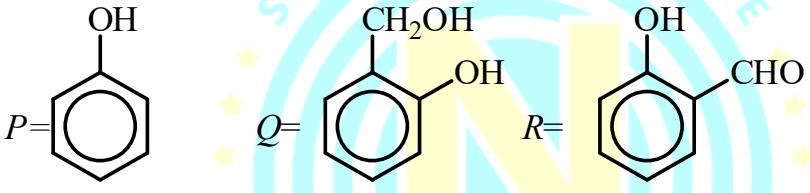
C) 

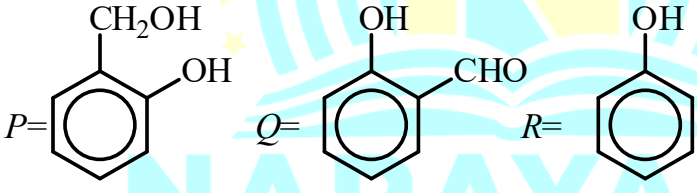
D) 

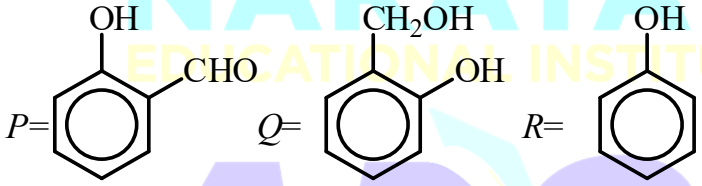
For More Material Join: @JEEAdvanced_2025

29. An organic compound $P(C_6H_6O)$ gives violate colouration with ferric chloride. On treatment with $CHCl_3$ and KOH, followed by acidification gives compound Q. Compound Q can also be obtained from compound R on reaction with pyridinium chlorochromate (PCC). Identify P, Q and R

A) 

B) 

C) 

D) 

30. Which of the following statement is incorrect?

A) Glucose give schiff's test and forms the hydrogen sulphite addition product with $NaHSO_3$

B) The pentacetate of glucose does not react with hydroxyl amine indicating the absence of free – CHO group

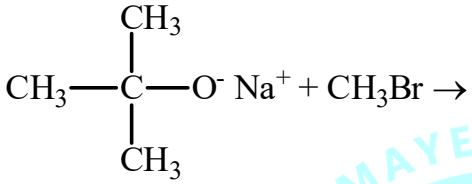
C) On oxidation with conc. HNO_3, both glucose and gluconic acid yield saccharic acid

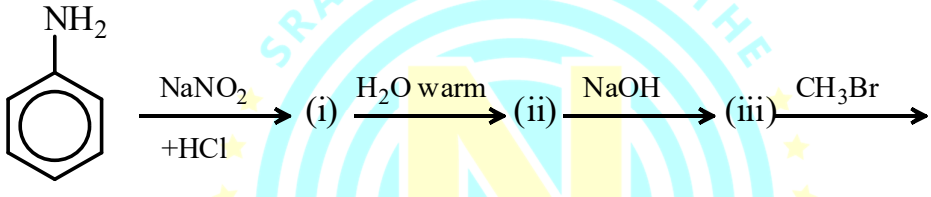
D) Glucose reacts with hydroxylamine to form an oxime

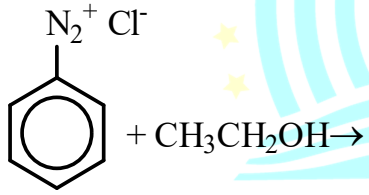
For More Material Join: @JEEAdvanced_2025

31. Which of the following doesn't produce an ether compound as a major product?

A) $CH_3CH_2OH \xrightarrow[140°C]{H_2SO_4}$

B) 

C) 

D) 

32. **Assertion:** Phenoxide ion is more stable than carboxylate ion hence phenol is more acidic than carboxylic acids.

Reason: The negative charge is delocalized over two electronegative oxygen atoms in carboxylate ion.

A) Both A and R are true and R is the correct explanation of A

B) Both A and R are true and R is not the correct explanation of A

C) A is true but R is false

D) A is false but R is true

For More Material Join: @JEEAdvanced_2025

33. Match the following.

List – I (Complex)		List-II (Hybridisation)	
a)	$\left[Ni(CO)_4\right]$	I.	sp^3
b)	$\left[Ni(CN)_4\right]^{2-}$	II.	sp^3d^2
c)	$\left[Co(CN)_6\right]^{3-}$	III.	d^2sp^3
d)	$[CoF_6]^{3-}$	IV.	dsp^2

A) $a \to IV, b \to I, c \to III, d \to II$ B) $a \to I, b \to IV, c \to III, d \to II$

C) $a \to I, b \to IV, c \to II, d \to III$ D) $a \to IV, b \to I, c \to II, d \to III$

34. Consider the complex ions, trans - $\left[Co(en)_2Cl_2\right]^+(A)$ and cis-$\left[Co(en)_2Cl_2\right]^+(B)$

The correct statement regarding them is

A) Both (A) and (B) can be optically active

B) Both (A) and (B) cannot be optically active

C) (A) can be optically active, but (B) cannot be optically active

D) (A) cannot be optically active, but (B) can be optically active

35. Which of the following electronic configuration of the corresponding element is wrongly matched?

A) Thulium - $[Xe]4f^{13}6s^2$ B) Americium - $[Rn]5f^77s^2$

C) Uranium - $[Rn]5f^47s^2$ D)Terbium - $[Xe]4f^96s^2$

36. Consider the ions/molecule $O_2^+, O_2, O_2^-, O_2^{2-}$ for increasing bond order the correct option.

A) $O_2^{2-} < O_2^- < O_2 < O_2^+$ B) $O_2^- < O_2^{2-} < O_2 < O_2^+$

C) $O_2 < O_2^{2-} < O_2^- < O_2^+$ D) $O_2^- < O_2^+ < O_2^{2-} < O_2$

For More Material Join: @JEEAdvanced_2025

37. Correct order of stability of halogen oxides

A) $I > Br > Cl$ B) $Br > Cl > I$ C) $Cl > Br > I$ D) $I > Cl > Br$

38. First and second ionization energies of magnesium are 7.646 and $15.035 eV$ respectively. The amount of energy in kJ needed to convert all the atoms of magnesium into Mg^{2+} ions present in $12\ mg$ of magnesium vapours is nearly:

[Given: $1eV = 96.5\ kJ\ mol^{-1}$]

A) 1.1 B) 1.5 C) 2.0 D) 0.5

39. 5 moles of PCl_5 and 2 moles of N_2 are placed in $200\ Lt$ vessel maintained at $600\ K$. The equilibrium pressure is $2.46\ atm$. The equilibrium constant (K_p) for the dissociation of PCl_5 is nearly _____ atm.

(Given $R = 0.082\ lit\ atm\ mol^{-1}\ K^{-1}$)

A) 3.3 B) 2.2 C) 1.1 D) 1.6

40. The freezing point of benzene decreases by $0.45°C$ when $0.2g$ of acetic acid is added to $20g$ of benzene. If acetic acid associates to form a dimer in benzene, percentage association of acetic acid in benzene will be: (K_f for benzene = $5.12\ K\ kg\ mol^{-1}$)

A) 74.6% B) 94.6% C) 64.6% D) 80.4%

41. Consider the following changes

1. Sublimation energy of $M = P$
2. Sublimation energy $+I.E_1 + I.E_2$ of $M = Q$
3. $I.E_1$ of $M = R$
4. $I.E_2$ of $M = S$
5. $I.E_1 + I.E_2$ of $M = T$

The enthalpy change of the reactions

(i) $M_{(s)} \rightarrow M^{+2}_{(g)} + 2e^-$,(ii) $M^{+}_{(g)} \rightarrow M^{+2}_{(g)} + e^-$ respectively are

A) (i) $Q - P + T$, (ii) $R + T$ B) (i) $P + R$, (ii) $T - R$

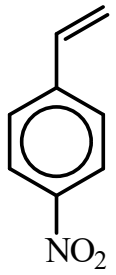
C) (i) Q, (ii) T D) (i) $P + R + S$, (ii) $R + S$

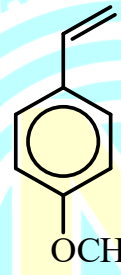
For More Material Join: @JEEAdvanced_2025

42. A solution of 15 mL contained $NaHCO_3$ and Na_2CO_3 mixture. It required 5 ml of $M/10$ HCl for neutralization using phenolphthalein indicator. To the same solution after adding methyl orange indicator it required more 15 mL of acid for neutralization. The amount of $NaHCO_3$ present in 500 mL of solution is

A) $2.8g$ B) $3.53g$ C) $5.6g$ D) $0.42g$

43. Arrange in decreasing order or reactivity with HCl

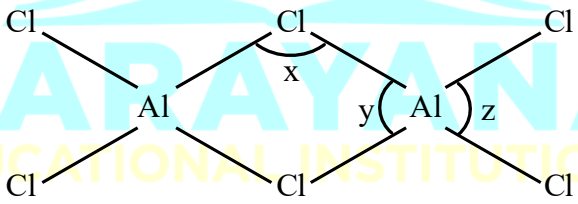
(i) 

(ii) 

(iii) 

A) $I > II > III$ B) $III > II > I$ C) $II > III > I$ D) $I > III > II$

44. Correct decreasing order of bond angle x, y and z in given structure.



A) $x > y = z$ B) $z > y > x$ C) $x > y > z$ D) $z > x > y$

45. Arrange the following in decreasing order of ionic size.

A) $Ce^{3+} > Lu^{3+} > Gd^{3+} > Eu^{3+}$ B) $Ce^{3+} > Eu^{3+} > Gd^{3+} > Lu^{3+}$

C) $Eu^{3+} > Ce^{3+} > Gd^{3+} > Lu^{3+}$ D) $Gd^{3+} > Eu^{3+} > Ce^{3+} > Lu^{3+}$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

For More Material Join: @JEEAdvanced_2025

46. The minimum pH required to prevent the precipitation of ZnS in a solution containing $0.01M$ $ZnCl_2$ and saturated with $0.10\,M$ H_2S is

(Given $K_{sp}(ZnS)=10^{-21}$, $Ka_1 \times Ka_2$ of $H_2S=10^{-20}$)

47. A tetrapeptide on hydrolysis produced glycine (gly), Leucine (Leu.), Alanine (Ala), Valine (Val.). Count number of possible tetrapeptides (structural isomers only) Which can produce above four amino acids with the following both conditions.

i) Glycine is not present at N-terminal

ii) Alanine in present at C-terminal

48. For a first order reaction $A \rightarrow B$. The rate constant $K=5.5\times10^{-14}\,\text{sec}^{-1}$. The time required for 67% completion of reaction is $x\times10^{-1}$ times the half life of reaction. Then 'x' (nearest integer) is _____ $(\log 3=0.48)$

49. $2MnO_4^- + bC_2O_4^{2-} + cH^+ \rightarrow xMn^{2+} + yCO_2 + zH_2O$

If the above equation is balanced with integer coefficients, the value of $(b+c)-(x+y+z)=$_____

50. For the elementary reaction $M \rightarrow N$, the rate of disappearance of M increase by a factor of 8 upon doubling the concentration of M. The order of the reaction with respect of M is

For More Material Join: @JEEAdvanced_2025

MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Given a real valued function $f(x)$, defined as $f(x)=\begin{cases}\dfrac{\tan^2\{x\}}{(x-[x])^2}, & x>0\\ 1, & x=0\\ \sqrt{\{x\}.\cot\{x\}}, & x<0\end{cases}$ where

$[x]$ and $\{x\}$ represents greatest integer function and fractional part of x, respectively, then which is incorrect?

A) $\lim\limits_{x\to 0^+} f(x)=1$ B) $\cot^{-1}\left(\lim\limits_{x\to 0^-} f(x)\right)^2=1$

C) $\tan^{-1}\left(\lim\limits_{x\to 0^+} f(x)\right)=\dfrac{\pi}{4}$ D) $\lim\limits_{x\to 0} f(x)$ exists

52. Let z be a complex number such that $\left|\dfrac{z-i}{z+2i}\right|=1$ and $|z|=\dfrac{5}{2}$. Then the value of $|z+3i|$ is

A) $\sqrt{10}$ B) $\dfrac{7}{2}$ C) $\dfrac{15}{4}$ D) $2\sqrt{3}$

53. A square is inscribed in the circle $x^2+y^2-10x-6y+30=0$. One side of this square is parallel to $y=x+3$. If (x_i,y_i) are the vertices of the square, then $\sum\left(x_i^2+y_i^2\right)$ is equal to

A) 148 B) 156 C) 160 D) 152

For More Material Join: @JEEAdvanced_2025

54. Four points having position vectors $\vec{a},\vec{b},\vec{c},\vec{d}$ are coplanar and $(\sin\alpha)\vec{a}+(2\sin 2\beta)\vec{b}+(3\sin 3\gamma)\vec{c}-\vec{d}=\vec{0}$, then the least value of $\sin^2\alpha+\sin^2 2\beta+\sin^2 3\gamma$ is

A) $\frac{1}{14}$ B) 14 C) 6 D) $\frac{1}{\sqrt{6}}$

55. If $f(x)$ satisfies the relation $f\left(\frac{5x-3y}{2}\right)=\frac{5f(x)-3f(y)}{2}, \forall x,y\in R$ and $f(0)=3$, $f'(0)=2$, then the fundamental period of $\sin(f(x))$ is

A) π B) 2π C) 3π D) 4π

56. Let P be the point $(10,-2,-1)$ and Q be the foot of the perpendicular drawn from the point $R(1,7,6)$ on the line passing through the points $(2,-5,11)$ and $(-6,7,-5)$. Then the length of the line segment PQ is equal to

A) 14 B) 13 C) 12 D) 17

57. The function $f(x)=\lim\limits_{x\to\infty}\frac{\cos(2\pi x)-x^{2n}\sin(x-1)}{1+x^{2n+1}-x^{2n}}$ is continuous for all x in

A) $R-\{-1\}$ B) $R-\{-1,1\}$ C) $R-\{1\}$ D) $R-\{0\}$

58. The total number of 3×3 matrices A having entries from the set $\{0,1,2,3\}$ such that the sum of all the diagonal entries of AA^T is 9, equal to ____

A) 765 B) 756 C) 766 D) 757

59. $\frac{1}{1!.100!}+\frac{1}{3!.98!}+\frac{1}{5!.96!}+\ldots+\frac{1}{(101)!}=$

A) $\frac{2^{100}}{100!}$ B) $\frac{2^{101}}{101!}$ C) $\frac{2^{100}}{(101)!}$ D) $\frac{2^{51}}{(101)!}$

60. If vertex of a parabola is $(2,-1)$ and the equation of its directrix is $4x-3y=21$, then the length of its latus rectum is

A) 2 B) 8 C) 12 D) 16

For More Material Join: @JEEAdvanced_2025

61. The general solution of the differential equation $x\left(y^2.e^{xy}+e^{\frac{x}{y}}\right)dy=y\left(e^{\frac{x}{y}}-y^2.e^{xy}\right)dx$ is

A) $e^{xy}=e^{\frac{x}{y}}+c$ B) $e^{xy}=e^{\frac{y}{x}}+c$ C) $e^{xy}+e^{\frac{x}{y}}=c$ D) $e^{xy}+e^{\frac{y}{x}}=c$

62. If from each of the three boxes containing 3 white and 1 black, 2 white and 2 black, 1 white and 3 black balls, one ball is drawn at random, then the probability that 2 white and 1 black ball will be drawn is

A) $\frac{13}{32}$ B) $\frac{1}{4}$ C) $\frac{1}{32}$ D) $\frac{3}{16}$

63. The integral $\int\frac{dx}{(x+4)^{8/7}(x-3)^{6/7}}$ is equal to (Where C is a constant of integration)

A) $\left(\frac{x-3}{x+4}\right)^{\frac{1}{7}}+C$ B) $-\left(\frac{x-3}{x+4}\right)^{-\frac{1}{7}}+C$ C) $\frac{1}{2}\left(\frac{x-3}{x+4}\right)^{\frac{3}{7}}+C$ D) $\frac{-1}{13}\left(\frac{x-3}{x+4}\right)^{\frac{-13}{7}}+C$

64. Let x denote the total number of one-one functions from set A with 4 elements to a set B with 5 elements and y denote the total number of one-one functions from the set A to the set $A\times B$ then

A) $y=969x$ B) $2y=91x$ C) $y=91x$ D) $2y=273x$

65. A ray of light through $(2,1)$ is reflected at a point P on the y-axis and then passes through the point $(5,3)$. If this reflected ray is the directrix of an ellipse with eccentricity $\frac{1}{3}$ and the distance of the near focus from the directrix is $\frac{16}{\sqrt{53}}$ then the equation of the directrix can be

A) $2x-7y-25=0$ B) $2x-7y+25=0$

C) $2x-7y-47=0$ D) $2x+7y-47=0$

For More Material Join: @JEEAdvanced_2025

66. Let $F:[3,5]\to R$ be a twice differential function on $(3,5)$ such that

$F(x)=e^{-x}\int_3^x\left(3t^2+2t+4F'(t)\right)dt$ Then $F(x)=$_____

A) $\frac{x^3-x^2-36}{e^x+4}$ B) $\frac{x^3+x^2-36}{e^x+4}$

C) $\frac{x^3+x^2-36}{e^x-4}$ D) $\frac{x^3+x^2+36}{e^x-4}$

67. Let $u=\frac{2z+i}{z-ki}, z=x+iy$ and $k>0$. If the curve represented by $\mathrm{Re}(u)+\mathrm{Im}(u)=1$ intersects the $y-$ axis at the points P and Q where $PQ=5$, then the value of k is:

A) $1/2$ B) $3/2$ C) 4 D) 2

68. The true solution set of $\log_{\tan\frac{\pi}{6}}\left(\frac{x^2-x+1}{x+2}\right)>-2$ is equal to

A) $-1<x<5$ B) $-2<x<5$

C) $-\infty<x<-2\ \cup\ -1<x<5$ D) $x\geq 5$

69. If the mean deviation of number $1,1+d,1+2d,...1+100d$ from their mean is 255, then d is equal to

A) 10.0 B) 20.0 C) 10.1 D) 20.2

70. Unit vector $\vec{c}$ is inclined at an angle θ to unit vectors $\vec{a}$ and $\vec{b}$ which are perpendicular. If $\vec{c}=\lambda\left(\vec{a}+\vec{b}\right)+\mu\left(\vec{a}\times\vec{b}\right)$ where λ,μ be real, then θ belongs to

A) $\left(-\frac{\pi}{4},0\right)$ B) $\left[0,\frac{\pi}{4}\right)$ C) $\left[\frac{3\pi}{4},\pi\right)$ D) $\left[\frac{\pi}{4},\frac{3\pi}{4}\right]$

SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

For More Material Join: @JEEAdvanced_2025

71. Let $f(x)=\begin{cases} xe^{ax} & , \quad x\le 0 \\ x+ax^2-x^3 & , \quad x>0 \end{cases}$, where '$a$' is a positive constant. The interval in which $f'(x)$ is increasing is $\left[\frac{k}{a},\frac{a}{l}\right]$. Then $k+l$ is equal to.

72. The value of $12\int_0^3 \left|x^2-3x+2\right| dx$ is

73. The area (in sq. units) of the region bounded by the curve $x^2=4y$ and the straight line $x=4y-2$ is k, then $8k$ is

74. For a unique value of μ & λ, the system of equations given by

$x+y+z=6$

$x+2y+3z=14$

$2x+5y+\lambda z=\mu$

has infinitely many solutions, then $\frac{\mu-\lambda}{4}$ is equal to

75. Four persons can hit a target correctly with probabilities $\frac{1}{2},\frac{1}{3},\frac{1}{4}$ and $\frac{1}{8}$ respectively. If all hit at the target independently, then the probability that the target would be hit, is n. Then $192n$ is _____

For More Material Join: @JEEAdvanced_2025



APEX JEE-MAIN 2025
For More Material Join: @JEEAdvanced_2025



APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025